Exercises 3.5 5, 7, 9, 11, 13, 15, 17, 19, 27, 35, 49, 51, 57, 63, 75.

5. $x^2 - 4xy + y^2 = 4$

$$2x - 4y - 4x\frac{dy}{dx} + 2y\frac{dy}{dx} = 0$$

$$(-4x + 2y)\frac{dy}{dx} = 4y - 2x$$

$$\frac{dy}{dx} = \frac{4y - 2x}{2y - 4x}$$

$$\frac{dy}{dx} = \frac{2y - x}{y - 2x}$$

7. $x^4 + x^2y^2 + y^3 = 5$

$$4x^3 + 2xy^2 + x^2 \cdot 2y\frac{dx}{dx} + 3y^2\cdot\frac{dy}{dx} = 0$$

$$(2x^2y + 3y^2)\frac{dy}{dx} = -4x^3 - 2xy^2$$

$$\frac{dy}{dx} = \frac{-4x^3 - 2xy^2}{2x^2y + 3y^2}$$

Exercises 3.5 5, 7, 9, 11, 13, 15, 17, 19, 27, 35, 49, 51, 57, 63, 75.

9. $\dfrac{x^2}{x+y} = y^2 + 1$

$x^2 = (y^2+1)(x+y)$

$x^2 = xy^2 + y^3 + x + y$

$2x = y^2 + x \cdot 2y\dfrac{dy}{dx} + 3y^2\dfrac{dy}{dx} + 1 + \dfrac{dy}{dx}$

$2x - y^2 - 1 = (2xy + 3y^2 + 1)\dfrac{dy}{dx}$

$\dfrac{dy}{dx} = \dfrac{2x - y^2 - 1}{2xy + 3y^2 + 1}$

11. $\sin x + \cos y = 2x - 3y$

$\cos x - \sin y \cdot \dfrac{dy}{dx} = 2 - 3\dfrac{dx}{dx}$

$(3 - \sin y)\dfrac{dy}{dx} = 2 - \cos x$

$\dfrac{dy}{dx} = \dfrac{2 - \cos x}{3 - \sin y}$

13. $\sin(x+y) = \cos x + \cos y$

$\cos(x+y)\left(1+\frac{dy}{dx}\right) = -\sin x - \sin y \cdot \frac{dy}{dx}$

$\left[\cos(x+y) + \sin y\right]\frac{dy}{dx} = -\sin x - \cos(x+y)$

$\frac{dy}{dx} = \frac{-\sin x - \cos(x+y)}{\cos(x+y) + \sin y}$

15. $y\cos x = x^2 + y^2$

$\frac{dy}{dx}\cos x - y\sin x = 2x + 2y\frac{dy}{dx}$

$(\cos x - 2y)\frac{dy}{dx} = 2x + y\sin x$

$\frac{dy}{dx} = \frac{2x + y\sin x}{\cos x - 2y}$

Exercises 3.5 5, 7, 9, 11, 13, 15, 17, 19, 27, 35, 49, 51, 57, 63, 75.

17. $2xe^y + ye^x = 3$

$$2e^y + 2xe^y \frac{dy}{dx} + \frac{dy}{dx} e^x + ye^x = 0$$

$$(2xe^y + e^x)\frac{dy}{dx} = -2e^y - ye^x$$

$$\frac{dy}{dx} = \frac{-2e^y - ye^x}{2xe^y + e^x}$$

19. $\sqrt{x+y} = x^4 + y^4$

$$\frac{1}{2}(x+y)^{-\frac{1}{2}}\left(1 + \frac{dy}{dx}\right) = 4x^3 + 4y^3\frac{dy}{dx}$$

$$\frac{1}{2\sqrt{x+y}} + \frac{1}{2\sqrt{x+y}}\frac{dy}{dx} = 4x^3 + 4y^3\frac{dy}{dx}$$

$$\left(\frac{1}{2\sqrt{x+y}} - 4y^3\right)\frac{dy}{dx} = 4x^3 - \frac{1}{2\sqrt{x+y}}$$

$$\frac{dy}{dx} = \frac{4x^3 - \frac{1}{2\sqrt{x+y}}}{\frac{1}{2\sqrt{x+y}} - 4}$$

Exercises 3.5 5, 7, 9, 11, 13, 15, 17, 19, 27, 35, 49, 51, 57, 63, 75.

27. $ye^{\sin x} = x\cos y$

$$\frac{dy}{dx}e^{\sin x} + y\cdot e^{\sin x}\cos x = \cos y - x\sin y\cdot\frac{dy}{dx}$$

$$\frac{dy}{dx}\left(e^{\sin x} + x\sin y\right) = \cos y - ye^{\sin x}\cos x$$

$$\frac{dy}{dx} = \frac{\cos y - ye^{\sin x}\cos x}{e^{\sin x} + x\sin y}$$

$$\left.\frac{dy}{dx}\right|_{\substack{x=0\\y=0}} = \frac{\cos(0) - 0\sin(0) - 0}{e^{\sin 0}}$$

$$= \frac{1-0}{1}$$

$$= 1$$

So the tangent line equation:

$$y - 0 = 1(x-0)$$

$$y = x$$

Exercises 3.5 5, 7, 9, 11, 13, 15, 17, 19, 27, 35, 49, 51, 57, 63, 75.

35. $2(x^2+y^2)^2 = 25(x^2-y^2)$

$2 \times 2(x^2+y^2)\left(2x+2y\frac{dy}{dx}\right) = 25\left(2x-2y\frac{dy}{dx}\right)$

$4(x^2+y^2)\left(2x+2y\frac{dy}{dx}\right) = 50x - 50y\frac{dy}{dx}$

$8x(x^2+y^2) + 8y(x^2+y^2)\frac{dy}{dx} = 50x - 50y\frac{dy}{dx}$

$(8x^2y + 8y^3 + 50y)\frac{dy}{dx} = 50x - 8x^3 - 8xy^2$

$\frac{dy}{dx} = \frac{50x - 8x^3 - 8xy^2}{50y + 8y^3 + 8x^2y}$

$\left.\frac{dy}{dx}\right|_{\substack{x=3\\y=1}} = \frac{50\times 3 - 8\times 3^3 - 8\times 3}{50 + 8 + 8\times 3^2} = \frac{-90}{130} = -0.692$

Exercises 3.5 5, 7, 9, 11, 13, 15, 17, 19, 27, 35, 49 ✓, 51, 57, 63, 75.

49. $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

$\frac{1}{a^2}(2x) - \frac{1}{b^2} \cdot (2yy') = 0$

$\frac{2}{a^2}x - \frac{2}{b^2}yy' = 0$ $\qquad \frac{2x}{a^2} \cdot \frac{b^2}{2y}$ $\frac{2y}{b^2}$

$\frac{2}{a^2}x = \frac{2}{b^2}yy'$

$y' = \frac{2b^2x}{2a^2y} = \frac{b^2x}{a^2y}$

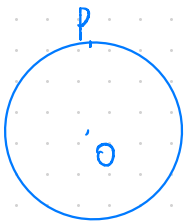
$y'(x_0, y_0) = \frac{2b^2x_0}{2a^2y_0}$

$y - y_0 = \frac{2b^2x_0}{2a^2y_0}(x - x_0)$

$y - y_0 = \frac{b^2x_0}{a^2y_0}(x - x_0)$

Exercises 3.5 5, 7, 9, 11, 13, 15, 17, 19, 27, 35, 49, 51, 57, 63, 75.

51.



$y'_{op} = \frac{1}{y'_p}$

let r be the radius of the circle

$x^2 + y^2 = r^2$

$2x + 2y \cdot y' = 0$

$y' = -\frac{x}{y}$

$m_1 m_2 = -1$

∴ Slope at $P_0(x_0, y_0)$ is $y'(x_0, y_0) = -\frac{x_0}{y_0}$

∵ op is perpendicular to the tangent line

∴ $y'_{op} = -\frac{1}{y'_{x_0 y_0}} = +\frac{y_0}{x_0}$

Exercises 3.5 5, 7, 9, 11, 13, 15, 17, 19, 27, 35, 49, 51, 57, 63, 75.

57. $\frac{1}{a^2} \cdot 2x + \frac{1}{b^2} \cdot 2y \cdot y' = 0$ orthogonal : $m_1 m_2 = -1$

$$\frac{2y}{b^2} \cdot y' = \frac{2x}{a^2}$$

$$y' = \frac{2b^2 x}{2a^2 y} = \frac{b^2 x}{a^2 y} = m_1$$

$$\frac{1}{A^2} \cdot 2x - \frac{1}{B^2} \cdot 2y \cdot y' = 0$$

$$-\frac{2y}{B^2} y' = -\frac{2}{A^2} x$$

$$y' = -\frac{2B^2}{2A^2 y} \cdot x$$

$$= -\frac{B^2 x}{A^2 y} = m_2$$

$$m_1 m_2 = \frac{b^2 x}{a^2 y} \cdot \left(-\frac{B^2 x}{A^2 y}\right) = -\frac{b^2 B^2}{a^2 A^2} \cdot \frac{x^2}{y^2}$$

(1, 1)

(-1, -1)

63. $x^2y^2 + xy = 2$

$$2xy^2 + x^2 \cdot 2y \cdot y' + y + xy' = 0$$

$$(2x^2y + x)y' = -2xy^2 - y$$

$$y' = \frac{-2xy^2 - y}{2x^2y + x}$$

$y' = -1$

$$\frac{-2xy^2 - y}{2x^2y + x} = -1$$

$$2xy^2 + y = +2x^2y + x$$

$$y(2xy + 1) = x(2xy + 1)$$

$$(2xy + 1)(y - x) = 0 \Rightarrow$$

$2xy + 1 = 0$ or

$xy = -\frac{1}{2}$

$y = x$

检验：

if $xy = -\frac{1}{2}$

$x^2y^2 + xy$

$= (-\frac{1}{2})^2 - \frac{1}{2} = -\frac{1}{4} \neq 2$

So, we must have $x = y$

$x^2y^2 + xy = 2$

$x^4 + x^2 = 2$

$x^4 + x^2 - 2 = 0$

$(x^2 + 2)(x^2 - 1) = 0$

$x^2 = -2$ (impossible)

$x = \pm 1$

$y = x$ $\Rightarrow$

75.